# Bags, Vectors & Transformers

Scaling Texts with Pairwise Comparisons · Bonus, Part 14

Denise J. Roth, PhD Candidate
Strategic Communication Group · Wageningen University & Research

# Categories or Degrees?

Many concepts are a matter of **degree**: how uncivil is a comment, how far left is a manifesto, how strong is an argument. A classifier forces them into boxes; a **scale** keeps the differences.

Ways to get a scale from text:

- dictionary counts (Day 1)
- human ratings, or human **comparisons**
- an LLM rating each text
- an LLM **comparing texts in pairs**

**Why compare?** Ask ten people to rate a comment from 0 to 9 and you get ten scales: one person's 6 is another's 8. Ask which of two comments is ruder, and they mostly agree. (Carlson & Montgomery, 2017)

**Content note.** The notebook uses real online comments, including insults and profanity.

# Two Ways to Ask the Model

**Direct rating** one prompt per text

```
system:  A toxic comment is rude,
disrespectful or unreasonable [...]
Rate the comment from 0 (not toxic
at all) to 9 (extremely toxic).
Reply with one digit.
user:    Comment:  {text}
```

We read the probability of each digit and take the **expected rating**: a model torn between 2 and 7 lands in between.

**Pairwise comparison** many prompts per text

```
system:  [...]  You will see two
comments, A and B. Which one
is more toxic?  Reply with A or B.
user:    Comment A: {text i}
Comment B: {text j}
```

We read $P(\text{A})$: a **soft win** for A. As on Day 4, the model never writes; we only read its probabilities, so every run gives the same result.

# Ask Both Ways: Position Bias

LLMs, like people, tend to prefer the **first** (or the second) option, whatever it says. So every pair is asked **twice**:

A = $i$, B = $j$ $\rightarrow$ $p_{ij}$

A = $j$, B = $i$ $\rightarrow$ $p_{ji}$

$$p = \tfrac{1}{2}\left(p_{ij} + 1 - p_{ji}\right)$$

If $i$ really is more toxic, $p_{ij}$ is high **and** $p_{ji}$ is low. If both are high, the position decided, not the content.

**Measure it and report it:**

- average probability for whichever comment came first (no bias: 0.50)
- share of pairs where the order, not the content, decided

Known from LLM-as-a-judge research (Zheng et al., 2023); small models are especially prone to it.

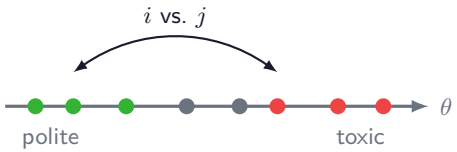
# From Comparisons to a Scale: Bradley–Terry

Every text gets a position $\theta$ on the scale (Bradley & Terry, 1952):

$$P(i \text{ judged more toxic than } j) = \frac{e^{\theta_i}}{e^{\theta_i} + e^{\theta_j}}$$

Like rating chess players from who beat whom.

- a few lines of code (Hunter, 2004)
- soft wins (0.8 = "probably $i$") count as fractions
- a small **prior** keeps a text that won every comparison from running off to infinity




**Design:** each round pairs every text with one other. 60 comments × 8 rounds = 240 pairs, × 2 orders = **480 prompts**.

# Data and Validation

**Civil Comments** (Borkan et al., 2019): comments from news sites' comment sections, 2015–2017. Each was shown to several crowd raters; **toxicity** is the **share of raters** who found it toxic.

Most comments aren't toxic, so we draw **10 comments from each of six toxicity bands**, 15–80 words, leaving out identity attacks, threats and sexual content.

**Validation**

- Spearman correlation of each scale with the crowd score
- agreement on pairs the raters clearly separate
- reading the comments where model and crowd disagree most

The crowd score is a measurement too: few raters per comment, and "toxic" meant different things to different raters.

# Cost, Reliability, and New Texts

**How many comparisons?**

Refit with 1, 2, . . . , 8 rounds and watch the correlation level off. One round costs as many prompts as the direct rating.

**How reliable?**

Split-half: a scale from the even rounds against one from the odd rounds. A measure can't agree with anything more than it agrees with itself.

**New texts?**

Compare each new text with ten **anchors** spread along the scale, and find its position with the anchors held fixed.

**For a large corpus:** scale a sample carefully, then anchor new texts, or **distill**: fine-tune a small model to predict the scale positions, as Day 4 did for categories.

# In the Notebook

- 60 Civil Comments across six toxicity bands
- Direct ratings: the expected digit from the model's probabilities
- 480 pairwise comparisons, in both orders, and the position bias
- Bradley–Terry in a few lines of NumPy
- Validation against the crowd, and reading the disagreements
- How many comparisons, split-half reliability, anchoring new comments

`bonus/notebooks/14_scaling_with_pairwise_comparisons_exercises.ipynb`
· 5 exercises · GPU